

BiliOnesie: A Phototherapy Garment for Treating Neonatal Jaundice

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Abstract

Neonatal jaundice is the most common medical condition in newborns, affecting more than 80% of infants. The common forms of phototherapy treatment for jaundice are uncomfortable for the infant and caregiver, disrupt important newborn support practices like holding the baby and early breastfeeding, and lose efficiency in shadowed body areas. Here, we present a phototherapy device in the form of a soft, stretchy, close-fitting full-body garment (a “onesie”). Our device uses surface-mount stitched e-textile fabrication methods to preserve the softness of the e-textile structure for next-to-skin applications. The prototype successfully produces 450-460 nm blue light at approximately 30 $\mu\text{W}/\text{cm}^2$ intensity at the garment surface, the recommended irradiance dose, and by nature of its close, conformable fit, it reaches more of the skin surface than traditional methods.

CCS Concepts

• Hardware ~ Emerging Technologies ~ Emerging Optical and Photonic Technologies • Hardware ~ Communication hardware, interfaces and storage ~ Sensors and Actuators

Keywords

E-textiles; wearable technology; jaundice; phototherapy

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1 Introduction

Hyperbilirubinemia, or jaundice, is a condition caused by excess bilirubin buildup in the body. Bilirubin is normally processed by the liver, but in utero, the bilirubin produced in the infant’s body is processed by the mother’s liver. Newborn livers are underdeveloped and often can’t process much bilirubin after birth. Further, bilirubin levels build up quickly in the newborn’s body as their red blood cells are rapidly replaced. As the most common medical condition in newborns (affecting more than 80% of infants [1]), jaundice treatment is a very common experience for newborns and caregivers.

Jaundice is treated by exposure to blue light, which helps the body break down bilirubin [2]. Current treatments for jaundice involve light therapy bassinets (bassinets that contain the baby while blue light is illuminated from above and/or below the bassinet, the most common treatment form) or blankets (flex LED circuit sheets, typically covered by a fabric layer, that can be wrapped around the infant) [3]. Bassinets are particularly traumatic for the infant and caregiver, as the newborn infant must be placed in the bassinet unclothed, wearing eye protection, for protracted periods. Newborns are accustomed to a warm, contained environment and are very uncomfortable under these conditions, which also causes distress for the caregiver. (Imagine a wailing newborn that the hormone-ridden caregiver is not allowed to pick up for comfort.) “Blanket”-form treatments are stiff, plastic devices that are uncomfortable and awkward to use. Treatment efficacy is maximized by the amount of surface area illuminated, as well as proximity to the light source [3]. Illumination from bassinets and blankets can’t reach into body folds and occluded areas, which limits bilirubin breakdown. Bassinets in particular also interfere with medically recommended early breastfeeding, holding the infant, and caregiver bonding in the crucial first days. Ironically, impaired breastfeeding exacerbates jaundice, since the infant’s first stool (meconium) contains high levels of bilirubin which can be reabsorbed by the intestines if it is not eliminated quickly [4]. Finally, all of these devices still require eye protection (which is also uncomfortable for the newborn). A few examples exist of garment-form phototherapy treatment devices. However, these examples are based on stiff plastic optical fiber [5], [6], limit the illumination area to a smaller portion of the body, and/or don’t account for stretch or conformability of the garment to the body surface [6], [7].

2 Garment Design

2.1 Design Objectives and Requirements

Phototherapy for jaundice treatment is most effective at a wavelength of 425-475 nm [3]. The dose/response relationship of phototherapy treatment is still being investigated, but current guidelines reflect an increase in efficacy as the dose increases, with an arbitrary limit currently established at $50\mu\text{W}/\text{cm}^2$. Overdose risks have not yet been studied effectively, so it is not known if exceeding $50\mu\text{W}/\text{cm}^2$ would be harmful. Existing dose/response studies have shown a plateau in effectiveness above about $30\mu\text{W}/\text{cm}^2$; therefore, that is the current irradiance target [2], [3].

As previously mentioned, surface area of exposure is closely linked to efficacy (as the light stimulus is received by the skin – therefore, the more skin exposed, the more bilirubin can be broken down). For this reason, the device should prioritize providing illumination to as much of the body as possible, including crevices and interiors of flexed joints, as well as the hands, feet, and head. Facial illumination is less desirable, as the infant's eyes should be protected from the light source.

Another essential consideration for the design of this kind of device is the newborn infant: promoting as much comfort as possible during treatment. Newborn skin is fragile; therefore all skin-contact surfaces should be soft, breathable, and hypoallergenic. Finally, it is important that a next-to-skin device does not pose any electrical shock hazards.



Figure 1: BiliOnesie garment design outside view, showing parallel traces to which interior surface-mount LEDs are soldered at a spacing of 2cm

2.2 Proof-of-Concept Prototype

The design of our garment is shown in Figure 1. The onesie covers the infant's head, hands, and feet (although LEDs are not embedded in the hand mitts for safety reasons). Serpentine sets of parallel traces are routed over the garment surface to which LEDs are soldered.

The light source is provided by a matrix of surface-mounted blue-light LEDs (Everlight Electronics, SMD 1113 package), which generate light at 450-460 nm (peak 455nm) according to the manufacturer's specifications. The LEDs are arranged in a parallel circuit that covers the garment surface with a spacing of 2cm between LEDs and 2mm between the stitched parallel power/ground traces. The LED circuit is embedded in the base fabric (cotton/Spandex jersey knit) using stitched traces (Syscom Liberator silver-coated Vectran thread) and soldered component joints based on the methods developed in [8]. Serpentine rows of parallel conductive traces were stitched to the textile surface using a Brother BAS-342G pattern stitching machine, and adjacent rows were electrically connected to form a serpentine parallel circuit after stitching (Figure 2). LEDs were soldered to traces on individual pieces prior to assembling the garment.



Figure 2: Stitched surface-mount LED circuit (Left, illuminated garment front piece shown; Right, outside of garment showing non-conductive side of stitched traces)

In total, the garment is designed to accommodate 162 LEDs (note that only the right side has been populated with LEDs for the purpose of the proof-of-concept, as shown in Figure 3). The LEDs are arranged to face inward toward the wearer's skin, with the conductive component of the stitched trace on the inside. A second layer of the base fabric is placed between the circuit and the infant's skin, which contributes to more equal light distribution, protects the infant's skin from any sharp edges of LED packages or solder joints, and prevents any electrical interference between the skin and the circuit.

The full-body onesie form was designed to be tight-fitting in order to ensure the closest possible contact between the light source and the skin and to help the garment conform to the body during joint movements rather than wrinkling away from the skin.

3 Evaluation

Our prototype was evaluated with the standard measurement approach for jaundice phototherapy, using a BiliMeter (Olympic Medical Corp.) in a darkened room. The measurement setup is shown in Figure 4. The garment was placed inside-out on the mannequin form to accommodate the BiliMeter probe.



Figure 3: Proof-of-concept prototype, with LEDs populated on the interior of the left side

The front garment section was powered using an adjustable benchtop power supply at a range of levels, and irradiance was measured at each level toward the limit of $30\mu\text{W}/\text{cm}^2$. At each voltage step, irradiance was measured in 3 or more locations on the active surface, and the measurements were averaged. Results are shown in Figure 5.

As shown in Figure 5, the prototype was able to reach approximately the target irradiance level of $30\mu\text{W}/\text{cm}^2$ (the maximum average irradiance recorded was $28.67\mu\text{W}/\text{cm}^2$). This demonstrates that the garment form is capable of providing the prescribed dose of phototherapy.



Figure 4: Test setup showing Bili-meter measurement on garment surface. Garment was tested inside-out to accommodate the test probe

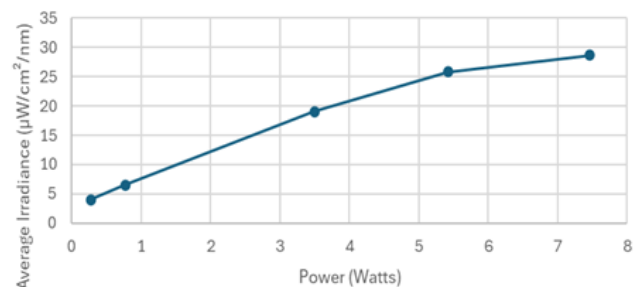


Figure 5: Relationship between irradiance and power for the front of the prototype phototherapy garment

4 Conclusion

This paper presents the development and initial evaluation of an e-textile phototherapy garment for treating neonatal jaundice. As shown in the evaluation results, the prototype is capable of providing the prescribed levels of illumination required for successful jaundice treatment. Its form factor overcomes many of the traditional challenges of phototherapy treatment: it is soft, warm, and comfortable for the newborn; allows the infant to be held and breastfed during treatment; illuminates more of the body's surface area than current treatments; and minimizes the

need for eye protection by bringing the light source onto the skin surface. The current proof-of-concept prototype has not yet been developed for deployment and would require optimization of the LED circuit, a user-friendly interface, and an on-board power source. However, in its current instance, the prototype demonstrates the feasibility of developing a soft, conformable, textile-based solution for phototherapy treatment.

Our future work involves characterizing the increase in treatment dose afforded by the garment design through a comparison of treatable surface area between our prototype and traditional methods; assessing the washability and reusability of the garment; and optimizing the circuit design, developing a user-friendly interface, and embedding an on-board power supply for the device prior to human testing.

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